GAGANYAAN MISSION

TOWARD INDIA'S FIRST HUMAN SPACEFLIGHT

All that has been done and what more needs to be done

LAUNCH VEHICLE

Human-rated LVM-3: For delivery of the Orbital Module into a 400 km low-earth orbit (LEO)

- Height: **43.4 m**
- Mass: **640 tonnes**
- Stages: **3 (solid, liquid, cryogenic)**
- Payload capacity: **10 tonnes**

STATUS: *Human-rating completed for all propulsion systems (i.e. certifying the vehicle is capable of safely transporting humans)*

CREW MODULE

Houses astronauts in Earth-like environment

Environmental Control & Life Support System (ECLSS)

- Maintains cabin pressure, humidity, temperature etc. in the module
- Maintains oxygen level, and removes carbon dioxide and trace contaminants

Cabin pressure: **98-116 kPa**

Humidity: **60-70%**

Temperature: **23±3°C**

Humancentric products

- Indigenous space suit
- Customised space food
- Ergonomically designed seating

STATUS: *Development/ testing in advanced stages*



CREW ESCAPE SYSTEM

Purpose

Rapid astronaut evacuation during launch emergencies

Components

- Solid motors for quick pull-away
- Autonomous activation for crew safety

STATUS: *Successful demonstration using test vehicle*

ORBITAL MODULE

It comprises two main parts

- **Crew Module (CM):** Houses astronauts in Earth-like environment

Mass: **5.3 tonnes**

- **Service Module (SM):** Supplies propulsion, power and thermal control to CM

Mass: **2.9 tonnes**

STATUS: *Development/ testing in advanced stages*

